# UNIVERSITY OF THE PUNJAB

B.S. in Computer Science Third Semester : Spring 2024 Roll ... [redacted]

Subject: Calculus and Analytical Geometry Paper: MS-251 N Time: 3 Hrs. Marks: 60

**THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED**

Q.1. Define the following: (6x5=30)

I. If $f(x) = x + 5$, $g(x) = x^2 - 3$. Find formula for $f(g(x))$. Also calculate $f(g(0))$?

II. 3 Evaluate $\lim_{x \to 1} \frac{x^2+x-2}{x^2-x}$?

III. Find an equation for the normal line to the curve at the given point. $x^2 - xy + y^2 = 7, (-1,2)$.

IV. Find the intervals of concavity and inflection points of $f(x) = x + \sin x$ for $0 \le x \le 2\pi$.

V. Evaluate the integral $\int \cos^2 x\, dx$?

VI. Find parametric quatioo for line through $P(-3,2,-3)$ and $Q(1,-1,4)$

0.2. Answer the following quations. (3110-30)

I. Find an equation for the tangent line to the curve at the given point. Then sketch the curve and tangent together. $f(x) = x^3, (-2,-8)$.

II. a) Find two positive numbers whose sum is 20 and whose product is as large as posible, b) Find all critical points of $f(x) = x^{4/3} - 4x^{1/3}$.

III. Find the point of intersection of the lines $L1: x = 1 + 2t, y = 2 + 3t, z = 3 + 4t$ by lines $L2: x = 2 + t, y = 4 + 2t, z = -1 - 4t$ and then find the plane determined lines?

Here's how to solve this problem step-by-step:

1. Find the formula for f(g(x))

The notation f(g@)) means we substitute the function g(x) into the function f (x). We are given that f@) = x -h 5 and g@) $= x^2 -$

To find f(g@)), we replace every instance of 'x' in the equation for f@) with the expression for g(x):

$f(g(x)) = (g(x)) + 5$

Now, substitute the expression for g@):

$f(g(x)) = (x^2 - 3) + 5$

Simplify the expression:

$f(g(x)) = x^2 + 2$

Therefore, the formula for f(g(x)) is c2 + 2.

2. Calculate f(g(O))

To calculate f(g(0)), we substitute a: O into the formula we just derived for $g(x$ f(g(0)) (0)2 +2

$f(g(0)) = 0 + 2$

f(g(0))$= 2$

Therefore, $f(g(0)) = 2.$

Here's how to solve this limit problem:

1. Factor the numerator and denominator:

The numerator is x2 + — 2. This factors to $(x+2)(x-1)$.

The denominator is x2 — x. This factors to — 1).

Therefore, the expression becomes: $\lim_{x \to 1} \frac{(x+2)(x-1)}{x(x-1)}$

2. Simplify the expression:

Notice that @ — 1) is a common factor in both the numerator and the denominator. We can cancel these factors, provided x # 1. Since we are taking the limit as approaches 1, this condition is satisfied. The simplified expression is:

lim $x+2$

3. Evaluate the limit:

Now, we can substitute x 1 into the simplified expression:

—

1

Therefore, the limit is 3.

The limit of the given expression as approaches 1 is 3.

Here's how to find the intervals of concavity and inflection points for the function f@) -k sin x on the interval 0 2m:

1. Find the first derivative:

The first derivative represents the slope of the function. We find it using the sum rule and the derivative of sin x:

$= \frac{d}{dx}(x + \sin x) = 1 + \cos m$

2. Find the second derivative:

The second derivative represents the rate of change of the slope, which determines concavity. We differentiate f'$x$):

$= \frac{d}{dx}(1 + \cos x) \quad - \sin x$

3. Find potential inflection points:

Inflection points occur where the second derivative changes sign. We set 0 and solve for x:

$- \sin x \; 0$

This equation is satisfied when 0, T, 2T. These are our potential inflection points.

4. Determine concavity intervals:

We test the intervals determined by the potential inflection points to see the sign of f"$(x)$:

- Interval (O, n): Choose a test point,

say $x = \frac{\pi}{2}$. Then
$f''(\frac{\pi}{2}) = -\sin(\frac{\pi}{2}) =$ —1 < 0.
Thus, the function is concave down on

- Interval (n, 2"): Choose a test point,

37T
say $x = \frac{3\pi}{2}$. Then
$f''(\frac{3\pi}{2}) = -\sin$ (—) = 1 > 0.
Thus, the function is concave up on (T, 2m).

5. Confirm inflection points:

Since the concavity changes at a: T, this is an inflection point. 0 and 27T are endpoints and do not represent a change in concavity within the given interval.

Final Answer: The function f (x) x + sin x is concave down on the interval (0, T) and concave up on the interval (T, 2m), with an inflection point at

Here's how to find the equation of the normal line to the curve a: xy + Y2 7 at the point (—1, 2):

1. Find the derivative:

We need to find using implicit differentiation. Differentiate both sides of the equation with respect to x:

$$2x - (y + x\frac{dy}{dx}) + 2y\frac{dy}{dx} = 0$$

2. Solve for dx.

Rearrange the equation to isolate:

$$2x - y - x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$$
$$(2y - x)\frac{dy}{dx} = y - 2x$$ (2y — x)

du _ y—2æ $dx$2y—x

3. Evaluate the derivative at the given point:

Substitute the coordinates of the given point (− $1, 2)$ into the expression for dc:

$\frac{dy}{dx}|_{(-1,2)} = \frac{2-2(-1)}{2(2)-(-1)} = \frac{4}{5}$

This is the slope of the tangent line at (−1, 2).

4. Find the slope of the normal line:

The normal line is perpendicular to the tangent line, so its slope is the negative reciprocal of the tangent line's slope:

mnormal $-\frac{1}{\phantom{0}}$ $\frac{5}{4}$

5. Use the point-slope form to find the equation of the normal line:

The point-slope form of a line is y − $y_1 = m(x - x_1$

, where (ml, Yl) is a point on the line and m is the slope. Using the point (−1, 2) and the slope

$m$ normal $-\frac{5}{\phantom{0}}$ we get:

$y-2=-\frac{5}{4}(x-(-1))$
$y-2=-\frac{5}{4}(x+1)$
$y-2=-\frac{5}{4}x-\frac{5}{4}$
$y=-\frac{5}{4}x-\frac{5}{4}+2$
$y=-\frac{5}{4}x+\frac{3}{4}$

The equation of the normal line to the curve at the point (−1, 2) is Y $=-\frac{5}{4}x+\frac{3}{4}$.

# Part 1: Evaluate the integral cos2x dx

To evaluate this integral, we use the powerreducing formula for cosine:

cos2a: = 1+cos(2æ) / 2

Substitute this into the integral:

f cos2:rdc f 1 +cos(2r,) dc

Now, we can integrate term by term:

$$= \frac{1}{2}\int(1+cos(2x))dx$$
$$= \frac{1}{2}\int 1dx+\frac{1}{2}\int cos(2x)dx$$

The integral of 1 is x, and the integral of cos(2x) is I sin(2.c). Therefore:

$$= \frac{1}{2}x+\frac{1}{4}sin(2x)+C$$

where C is the constant of integration.

## Part 2: Find parametric equations for the line through p(-3, 2, -3) and Q(l, -1, 4)

To find the parametric equations of a line passin through two points $P(x_1, y_1,$ Zl) and Q(C2, $y_2, z_2)$ we first find the direction vector 13'.

$$\vec{v} = \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \\ z_2 - z_1 \end{pmatrix} = \begin{pmatrix} 1-(-3) \\ -1-2 \\ 4-(-3) \end{pmatrix} = \begin{pmatrix} \\ \\ \\ \end{pmatrix}$$

4

-3

7

Now, we can write the parametric equations usim point P and the direction vector F:

$$x = x_1 + v_x t = -3 + 4t$$
$$y = y_1 + v_y t = 2 - 3t$$
$$z = z_1 + v_z t = -3 + 7t$$

where t is a parameter.

Therefore, the parametric equations for the line a $x =$ —3 + 4tj Y 2 — 3t, and z — —3 + 7t. The integral of cos2x is + åsin(2x) +

C and the parametric equations for the line are x — -3 + 4t, 2 - 3t z= -3 +

Here's how to solve this problem step-by-step:

1. Find the derivative:

The derivative of f@) x $^{3}$ is found using the power rule: f'(x) 3c $^{2}$ . This represents the slope of the tangent line at any point on the curve.

2. Find the slope at the given point:

The given point is (-2, -8). Substitute x — -2 into the derivative to find the slope of the tangent line at this point:

$m = f'(-2) = 3(-2)^2 =$ 3(4) 12

3. Use the point-slope form to find the equation of the tangent line:

The point-slope form of a line is y — $y_1 = m(x = x_1$

, where (ml, Yl) is a point on the line and m is the slope. Using the point (-2, -8) and the slope m = 12, we get:

$y - (-8) = 12(x - (-2))$

$y + 8 = 12(x + 2)$

y +8 — 12m +
24 y + 16

4. Sketch the curve and tangent line:

To sketch the curve f@) x3, plot a few points. It's a cubic function that passes through the origin (0,0) and increases rapidly as x moves away from O. The tangent line y 12m + 16 has a y-intercept of 16 and a slope of 12. Draw the tangent line touching the curve at the point (-2, -8).

The equation for the tangent line to the curve $) = x^3$ at the point (-2, -8) is y — 12T + 16.

Here's how to solve these problems:

a) Finding two positive numbers

Let the two positive numbers be and Y. We are given that their sum is 20, so we can write:

$+ y = 20$

We want to maximize their product, P xy. we can express Y in terms of using the sum equation:

y — 20 — $x$

Now substitute this into the product equation:

$(x) =$ x(20 − x) 20m − x $^2$

To find the maximum, we take the derivative of P (x) with respect to and set it to zero:

20 - = O

$\frac{}{dc} =$

Solving for c:

$2x =$ 20 a:

10

Now we find Y:

y = 20 $- x = 20$- 10 10

Therefore, the two positive numbers are 10 and 10. Their product is 10 x 10 100.

To confirm this is a maximum, we can take the second derivative:

d2P —2

dx2 —

Since the second derivative is negative, this confirms that we have a maximum.

The two positive numbers whose sum is 20 and whose product is as large as possible are 10 and 10.

b) Finding critical points

To find the critical points of f(x) — $x^{4/3} - 4x^{1/3}$, weneed to find where the derivative is zero or undefined. First, let's find the derivative:

$f'(x) = \frac{4}{3}x^{1/3} -$ 4x-2/3
3

Now we set the derivative equal to zero:

$x^{1/3}$ —4x¯2/3 O
3

$x^{1/3}$ =4x-2/3

x1/3 x-2/3

Cl/3 _ 2/31

Now let's check where the derivative is undefined. The derivative is undefined when a: 0 because of the x-2/3 term.

Therefore, the critical points of f@) are O and

The critical points of f@) æ4/3 — 4x1/3 are a: — O and a; = 1.

Here's how to solve this problem step-by-step:

1. Find the point of intersection:

To find the point of intersection of lines Ll and L2, we need to find values of t and s such that the x, Y, and z coordinates are equal for both lines. We set up a system of equations:

$1 + 2t = 2 + s$
$2 + 3t = 4 + 2s$
$3 + 4t = -1 - 4s$

We can solve this system of equations. Let's start by solving the first equation for s:

s 2t — 1

Now substitute this into the second equation:

$2 + 3t = 4 + 2(2t - 1)$
$2 + 3t = 4 + 4t - 2$
$2 + 3t = 2 + 4t$
$t = 0$

Now substitute t O back into the

equation for s: s — 2(0) — 1 —1

Finally, let's check if these values satisfy the third equation:

$3 + 4(0) = -1 - 4(-1)$

3—3

The values t 0 and s — —1 satisfy all three equations. Now substitute t — 0 into the equation for Ll (or s — —1 into the equation for L2) to find the point of intersection:

$x = 1 + 2(0) = 1$

$y = 2 + 3(0) = 2$

$z = 3 + 4(0) = 3$

The point of intersection is (1, 2, 3).

2. Find the plane determined by the lines:

To find the plane, we need two vectors that lie in the plane. we can use the direction vectors of the lines Ll and L2:

$\vec{v_1} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$ ( 3 (direction vector of Ll)

$\vec{v_2} = \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix}$ (direction vector of L2)

The normal vector to the plane is given by the cross product of these two vectors:

$$\vec{n} = \vec{v_1} \times \vec{v_2} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \times \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} = \begin{pmatrix} (3)(-4) - (4)(2) \\ (4)(1) - (2)(-4) \\ (2)(2) - (3)(1) \end{pmatrix}$$

Now we use the point of intersection (1, 2, 3) and the normal vector $\begin{pmatrix} -20 \\ 12 \\ 1 \end{pmatrix}$ to find the equation of the plane:

$-20(x$– I) 12(y — 2) -4— I (z$- 3) =$

−20m + 20 + 12!] — 24 -+- z – 3$= 0$

−20m $+ 12y +$z – 7$= 0$

$20x$- 12y -z +7=0

The plane determined by these lines is $20x$- 12y -z +7 = O.

Final Answer: The final answer is

20m - 12y -z +7=0